Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Physical AI-based Human-to-Humanoid Knowledge and Skill Transfer Framework for Automated Industrial Equipment Operation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical & Materials		Mechanical & Materials	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Robotics / Automation Machinery	Robot Control and Intelligence Technologies	Robotics/ Automation Machinery	Robot Vision and Production Automation Technologies
Concept of Proposed Technology	Avoidable productivity losses, safety risks, and labor shortages remain persistent challenges across field-based industries, including construction, manufacturing, and logistics. Robotic automation of industrial equipment operations like trucks, forklifts, loaders, and cranes can mitigate these challenges by improving productivity and reducing worker exposure to high-risk equipment-centered tasks and operating zones. However, scaling automation remains a critical bottleneck: primary approaches—such as retrofitting sensors onto each legacy machine or developing entirely new robotic platforms—are inherently time-consuming and difficult to generalize across diverse equipment fleets. Recent advances in humanoid robots, designed for human-level physical dexterity, hold the potential			

	to enable deployment within existing work infrastructure with minimal renovation. Yet, despite this potential, enabling humanoid robots to reliably operate complex industrial machinery remains largely underexplored, especially under field conditions that demand machine- and operation-specific expertise. Addressing this gap requires a control framework that can translate human expert intent into high-precision, robust physical execution. In this project, we aim to develop a **Physical AI framework that transfers human expert intent and control strategies to humanoid robots, enabling autonomous operation of heavy industrial equipment**—including legacy machines such as trucks, forklifts, loaders, and cranes. As an early attempt, we propose to demonstrate the proposed framework within a high-fidelity simulation environment featuring 6-DOF (Degrees of Freedom) industrial equipment interfaces. The findings of this project are expected to serve as a solid foundation for transitioning humanoid-led industrial equipment operations to real-world environments. The central concept is to model how expert operators determine their next actions by aligning cognitive intent signals—captured through EEG and eye-tracking–based visual and auditory attention—with time-synchronized equipment control logs. EEG-derived signals are treated as auxiliary intent cues that complement gaze- and control-log–based intent modeling, rather than serving as a primary control input. The resulting intent-aware representations are then used to learn robust whole-body control policies that enable humanoid robots to operate machinery interfaces under high-fidelity simulated conditions with precision and stability. Key features: ● **Experience-Informed Knowledge Transfer:** Encodes expert equipment-operation expertise by synchronizing EEG- and gaze-derived intent/attention cues with equipment control logs (e.g., steering angle, throttle/brake input, gear selection, velocity, and trajectory) to learn intuitive and context-dependent operation policies. ● **Cognitive-Visual and Auditory Integrated Perception:** Couples perception and reasoning with human-like attention priors to reliably detect hazards and identify navigation landmarks in cluttered, dynamic construction environments, improving situational awareness and decision reliability. ● **Embodied Physical AI for Complex Equipment Manipulation:** Translates intent-aware policies into stable whole-body control that coordinates arms, legs, and balance to execute precise steering, shifting, and pedal actions under unstructured terrain and variable contact conditions.
Rationale for Joint R&D Project	The proposed project requires close, continuous collaboration between a research institute and an industrial partner to successfully translate intent-aware Physical AI and whole-body humanoid control into a robust capability for autonomous operation of legacy industrial machinery. A joint R&D framework is essential to integrate advanced AI models with physical humanoid hardware and machine interfaces, and to safely validate the technology under high-fidelity simulated environments that rigorously replicate the safety-critical constraints of actual industrial sites. Here, simulated environments refer to a physical 6DOF motion-based simulator integrated with actual machine interfaces (steering, levers, pedals) that replicates the dynamic haptic feedback and safety-critical constraints of real-world equipment operation. Rationale for collaboration: ● The research institute provides core expertise in Physical AI, including intent modeling from EEG and eye tracking, multimodal perception and reasoning under cluttered environments, and contact-rich whole-body control for humanoid manipulation.

	● The industrial partner provides the domain-specific assets required for high-fidelity replication, including access to legacy equipment for interface instrumentation, real-world operator control logs, safety protocols, and technical specifications for digital-twin modeling. ● Effective co-design of AI software, humanoid hardware, and machine interfaces is necessary to meet stringent constraints on safety, reliability, latency, and fail-safe operation, which cannot be adequately addressed in simulation-only research. Expected benefits of joint R&D: ● Faster progression from abstract research concepts to integrated, lab-tested prototypes by incorporating real-world equipment specifications and operational constraints. ● Lower technical risk by continuously validating AI algorithms against actual equipment dynamics, operator workflows, and safety requirements, enabling early detection of failure modes within controlled environments. ● Development of a physics-grounded, reusable technology platform that can be scaled to multiple types of humanoids, legacy machinery, and diverse industrial scenarios in future R&D phases. ● Minimizing the discrepancy between virtual models and physical reality by utilizing the industrial partner's assets (e.g., machine control logs, interface data) for high-fidelity system modeling.
Key Words	1. Human Intent-Aware Physical AI 2. Human-to-Humanoid Knowledge and Skill Transfer 3. Humanoid Contact-Rich Whole-Body Control 4. Cognitive-Visual Integrated Perception 5. Autonomous Industrial Equipment Operation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (2)	End of the Project	Level (4)
R&D Objectives	The objective of this R&D project is to develop and laboratory-validate an intent-aware Physical AI framework that enables humanoid robots to learn and execute human expert intent and control strategies for operating representative legacy industrial equipment interfaces (e.g., steering, levers, pedals) under controlled, lab-based simulated conditions. The framework will integrate multimodal intent understanding (EEG, eye tracking), cognitive–visual and auditory integrated perception, and contact-rich whole-body control, and will be evaluated in a laboratory testbed—including a 6-DOF motion-based simulator and instrumented equipment interfaces—to support component-level and integrated-breadboard validation across a limited set of representative operation scenarios. This TRL-4 effort will establish the technical feasibility and key performance baselines needed to progress toward higher-fidelity demonstrations and future real-world deployment. The specific R&D objectives are: ● To develop and experimentally validate multimodal intent modeling and knowledge/skill transfer methods that synchronize EEG- and gaze-derived intent/attention cues with equipment control logs (e.g., steering, throttle/brake, gear state) to learn initial context-dependent operating policies in controlled laboratory scenarios. ● To develop and bench-test cognitive–visual and auditory integrated perception and reasoning that leverages human-like attention patterns to detect hazards and identify navigation landmarks in representative, lab-configured simulated scenes, establishing baseline robustness under controlled variability (e.g., clutter level, lighting, audio events). ● To develop and laboratory-validate embodied Physical AI control that enables a humanoid robot to execute stable, precise whole-body coordination for operating legacy-style machine interfaces (e.g., steering, shifting, pedal control) under controlled contact conditions, demonstrating feasibility and identifying key failure modes and control limits.			
R&D Target Specifications	The target performance will be evaluated in a controlled laboratory setting using a 6-DOF motion-based simulator and instrumented equipment interfaces. These specifications are intended as initial, lab-validation targets to establish feasibility and baseline performance (TRL 4), rather than field-level autonomy or robustness. The proposed target specifications of this R&D project are as follows: ● **Intent-Aware Knowledge and Skill Transfer Performance (≥ 0.7 correlation):** Achieve strong agreement between the learned intent-aware policy outputs and expert operation logs, with Spearman/Pearson correlation ≥ 0.7 across held-out scenarios of equipment operation. ● **Cognitive–Visual and Auditory Integrated Perception Performance (F1 ≥ 0.8):** Achieve F1-score ≥ 0.8 for detecting predefined site hazards and identifying navigation landmarks under clutter and occlusion in simulated dynamic industrial environments. ● **Embodied Equipment Operation Performance (≥ 80% success rate):** Achieve an operation success rate ≥ 80% for key legacy interface actions (e.g., steering to target, gear selection, pedal control) executed by a humanoid robot under simulated unstructured-terrain and variable contact conditions with variable contact forces, roll/tilt angles, and lateral g-force disturbances.			

	● **Safety Performance (collision-free operation; low intervention):** Demonstrate collision-free performance under predefined laboratory-scale, high-fidelity simulation scenarios, with minimal human intervention during structured evaluation tests.
R&D Plan	This R&D project will be carried out over three years, leveraging our existing high-fidelity 6-DOF motion-based simulator and our current humanoid robot platform to enable a progressive workflow—from expert data acquisition and accelerated virtual learning to integrated laboratory validation within the simulator environment. The project will utilize our **in-house 6-DOF simulator cockpit and the Unitree Robotics H1-2 humanoid robot** as the primary embodied system for developing and evaluating intent-aware Physical AI capabilities. ● Year 1: Multimodal Expert Data Acquisition and Skill Encoding ○ Recruit expert equipment operators and secure required human-subject approvals and safety protocols (e.g., IRB, risk controls). ○ Conduct standardized driving/operation sessions on a 6DOF motion-based simulator to collect expert operation logs across representative scenarios. ○ Build a synchronized multimodal dataset by time-aligning sensor-derived human intent signals—captured through wearable high-fidelity EEG, eye-tracking (gaze), and body-mounted IMUs—with spatial audio and vehicle control telemetry (e.g., steering angle, throttle/brake inputs, gear state, speed, and trajectory). ○ Develop intent-aware skill representations and benchmarks by training a neural–behavioral encoding model and defining hazard/landmark recognition labels grounded in expert visual attention. ● Year 2: AI-Driven Policy Training via High-Fidelity Simulation ○ Develop a digital twin of the industrial equipment and environment using the NVIDIA Isaac Sim platform. ○ Implement parallel reinforcement learning in a virtual environment to optimize the Intent-Aware Physical AI policy based on collected experts' data. ○ Train embodied whole-body control algorithms for precise manipulation of legacy interfaces (steering, pedals, levers) under simulated physical constraints. ○ Validate baseline performance and sensitivity of AI policies through Domain Randomization, accounting for various lighting, noise, weather, and terrain conditions. ● Year 3: Integrated Physical Validation and Embodied Intelligence Proof ○ Integrate the trained AI policy into the Unitree H1-2 humanoid robot seated within a 6DOF simulator cabin equipped with actual legacy interfaces. ○ Conduct Embodied Physical AI validation to assess the robot's ability to handle real-world physical resistance (haptics) and motion feedback. ○ Perform comparative analysis between the robot's policy execution in supervised evaluation tests and the Year 1 expert baseline to verify Knowledge and Skill Transfer performance. ○ Finalize the TRL 4 validation by demonstrating stable equipment operation and

	collision-free performance under predefined simulation scenarios. ○ (Optional case study) Conduct a limited, supervised trial on representative equipment in a controlled zone (e.g., low-speed, geofenced, E-stop-enabled) to document practical integration constraints and transferability beyond the simulator.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for developing the core Physical AI algorithms and AI models (humanoid control policy) trained on expert data, providing the system-level intelligence required to transfer human expert equipment-operation skills to humanoid robots and enable autonomous operation of heavy industrial equipment, including legacy machinery. Based on its experience in multimodal human sensing, computer vision, and robotic manipulation [1-5], the institute will lead the research on multimodal intent understanding (EEG, gaze, audio, and IMU-based body motion across legs, arms, and head), cognitive-visual and auditory integrated perception, decision-making, and embodied whole-body control needed for reliable performance within high-fidelity, physics-grounded laboratory environments that replicate dynamic jobsite conditions using field-derived realism inputs from the Korean SME partner. The specific roles and responsibilities of the research institute include: ● Expert Data Acquisition and Subject Recruitment: Leading the recruitment of expert heavy equipment operators and managing human subjects for high-quality data collection, ensuring the capture of professional-grade operational nuances. ● Development of intent-aware knowledge and skill transfer algorithms, including multimodal modeling that synchronizes EEG/eye-tracking with equipment control telemetry to learn expert operating strategies. ● Development of cognitive–visual and auditory integrated perception and reasoning, including hazard detection and landmark-based navigation that leverages human attention patterns for robust operation in cluttered, simulated industrial environments. ● Development and validation of embodied humanoid control, including physics-grounded whole-body policies for contact-rich manipulation of legacy interfaces (steering, pedals, levers, shifting) and safety-aware execution. ● Simulation-based development and evaluation, including digital-twin modeling, virtual accelerated learning (e.g., Isaac Sim), robustness testing (domain randomization), and rigorous performance/safety analysis using a 6DOF motion-based physical simulator. The institute will also provide ongoing technical leadership and co-design support with the industrial partner for system integration, safety protocols, and experimental validation.
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the Field Intelligence and Industrial Deployment Partner, ensuring the Physical AI framework is grounded in real-world industrial constraints and aligned with deployable workflows for legacy equipment operation. The SME will bridge core R&D and industrial application by providing field-derived requirements and data, coordinating access to expert operators and equipment, and leading integration constraints definition and industrialization planning for future field adoption. The specific roles and responsibilities of the Korean SME partner include: ● Field-Derived Requirements & Constraints Capture: Provide field-derived operational requirements and constraints for legacy equipment operation (e.g., typical task workflows, operating envelopes, safety practices, maintenance considerations, and site logistics) to guide system specifications. ● Realism Inputs for High-Fidelity Test Conditions: Supply jobsite-informed parameters to define and calibrate field-relevant disturbance conditions for laboratory testing (e.g., representative terrain/grade profiles, vibration signatures, roll/tilt ranges, contact variability, and environmental noise sources). This enables the research team to construct high-fidelity,

	physics-grounded test conditions that reflect real deployment contexts. ● Expert Operator Access & Structured Feedback: Coordinate access to veteran equipment operators and facilitate structured feedback on operational realism, interface interaction practicality, and expected professional operating standards. The SME will consolidate this feedback into actionable requirement updates and acceptance criteria definitions. ● Legacy Equipment Interface Characterization & Compatibility Guidance: Provide interface-level artifacts for representative legacy equipment (e.g., control geometries, force/torque ranges, operational procedures, photos/videos, and CAD/measurements as needed) and advise on interface variability across models and non-invasive compatibility considerations to support generalizable manipulation strategies. ● Deployment and Industrialization Roadmap Planning: Lead early planning to identify key integration requirements and constraints for moving beyond the lab (e.g., packaging considerations, environmental assumptions, safety requirements, serviceability, and cost drivers), producing a preliminary engineering roadmap to inform future pilots and higher-TRL efforts.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver a **Physical AI-driven humanoid control framework** capable of transferring human expert skills to autonomously operate legacy industrial equipment within high-fidelity, motion-based laboratory environments. The project will advance the state of the art in intent-aware skill encoding, cognitive-visual and auditory integrated perception, and embodied whole-body control, achieving measurable performance and safety improvements for humanoid manipulation of industrial machinery. From an industrial perspective, this project will provide the Korean SME partner with core **Physical AI-enabled robotic technologies** and system-integration know-how applicable to automating legacy heavy machinery without extensive vehicle modifications. The outcomes will serve as a foundational technical asset for future commercialization, enabling cost-effective and scalable robotic solutions for high-risk industrial sites. More broadly, the project is expected to: ● Lay the technical foundation to mitigate the critical shortage of skilled operators by providing a validated humanoid-based automation framework that can be scaled to real-world deployment in subsequent phases. ● Demonstrate the feasibility of augmenting human operators in high-risk environments, thereby establishing the safety protocols and control reliability necessary for future hazardous jobsite applications. ● Empower the Korean robotics SME with global-level AI expertise and high-fidelity physical validation capabilities, enabling them to lead the transition from simulation to industrial-scale implementation. ● Contribute to the expansion of the industrial robotics industry by proving the practical utility of embodied intelligence within rigorously simulated relevant environments, reducing the R&D barrier for future Physical AI applications.

6. References

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